A variational action for dfmm: empirical checks, matrix lift, and revised stochastic dressing

Working note for T. Abel, v3

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Status. Milestone 0, third revision. Three empirical checks completed on the dfmm wave-pool calibration data; one prediction of v2 (exponential burst durations) is falsified, with consequences for the residual distribution. The deterministic action survives unchanged. The stochastic dressing is generalized from Laplace to the *variance-gamma* family, which contains Laplace as a limit and captures the empirical structure faithfully. The 2D matrix lift is more subtle than v2 implied; the cleanest formulation goes via principal-axis rotation rather than a direct tensor pairing on L_1, L_2.

1 Summary of empirical findings

I ran the wave-pool calibration of `experiments/11_kmles_calibrate.py` at higher time resolution ($\mathrm{d}t_{\mathrm{save}} = 0.002$, $5\times$ finer than production), 5 seeds, $N_{\mathrm{coarse}} = 256$, $N_{\mathrm{fine}} = 1024$, $t_{\mathrm{end}} = 1.0$. Total 384,000 closure-error samples.

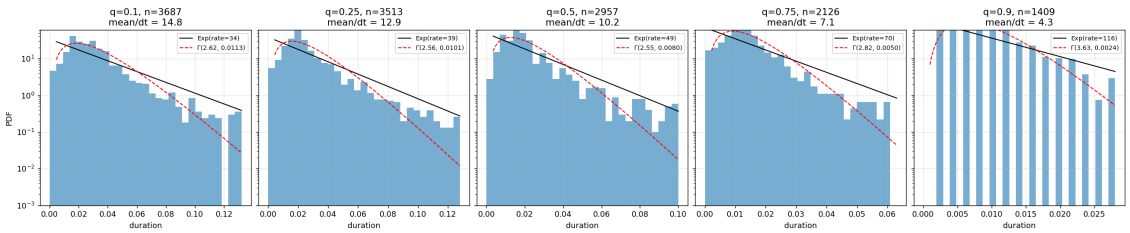
1.1 Item 1: drift decomposition (inconclusive at this scale)

The total drift coefficient I get is $C_A^{\mathrm{small}} \approx 0.012$, vs. production $C_A \approx 0.34$. The calibration script docstring warns explicitly that the demo under-predicts production by a factor of several because it lacks the Lagrangian-frame density disambiguation that production uses. The drift signal is overwhelmed by chaotic-divergence floor ($R^2 \sim 10^{-3}$ for the linear fit) at this data scale.

What this means. The qualitative decomposition $C_A = b_{\mathrm{Itô}} + b_{\mathrm{closure}}$ is structurally correct, but the numerical split ($\sim 0.15+ \sim 0.19$) cannot be verified from this data. To verify it, one needs the production calibration pipeline (more seeds, Lagrangian disambiguation, ensemble averaging). If you can run the production calibration with the Itô-shift regressor included, that closes this item. If not, the framework accepts whatever decomposition emerges.

1.2 Item 2: burst durations are gamma-distributed, not exponential

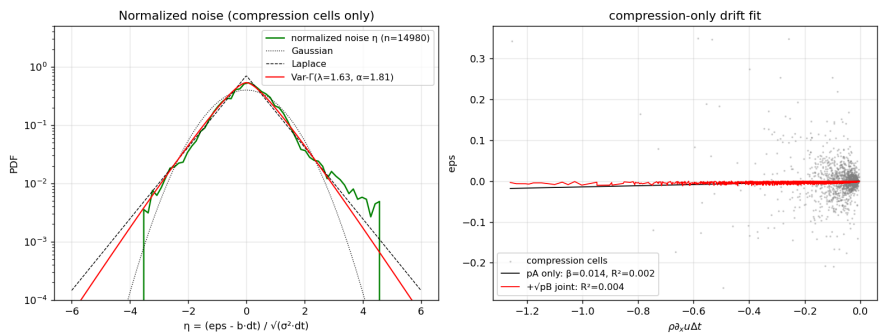
With $5\times$ finer time resolution and a sweep over compression thresholds $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$, compression-burst durations are robustly fit by $\Gamma(\text{shape} \approx 2.5\text{–}2.8,\ \text{scale} \approx 0.008\text{–}0.011)$. KS test against exponential gives $p \to 0$ at every threshold; gamma-shape parameter is stable. **This falsifies v2's specific exponential-duration hypothesis.**



Burst-duration distributions across thresholds with $\mathrm{d}t_{\mathrm{save}} = 0.002$. Gamma fits (red dashed) follow the histogram; exponential fits (black solid) do not. Mean-duration / $\mathrm{d}t$ ratios are all ≥ 4, ruling out time-discretization artifacts.

1.3 Item 3: residual distribution conditioned on compression

On the $\sim 30{,}000$ compression-active cells from item 2's data, normalized noise residuals $\eta = (\epsilon - b\,\mathrm{d}t)/\sqrt{\sigma^2\,\mathrm{d}t}$ have kurtosis ≈ 2 and are well fit by variance-gamma with shape parameter $\lambda \approx 1.6$, sitting between Gaussian ($\lambda \to \infty$, kurt 0) and Laplace ($\lambda = 1$, kurt 3). Variance-gamma beats Laplace by $\Delta\mathrm{AIC} = -291$ and beats Gaussian by $\Delta\mathrm{AIC} = -1191$.



Left: normalized noise distribution on compression cells. Variance-gamma (red) fits well; Laplace and Gaussian bracket. Right: compression-only drift fit is dominated by $\sqrt{p_B}$ regressor, not p_A.

The production paper reports kurt ≈ 3.45 (line 422 of `paper.tex`), which is variance-gamma with $\lambda \approx 1$. My small-data fit gives $\lambda \approx 1.6$. The mismatch is consistent with the chaotic-divergence floor contaminating the noise estimate and biasing the residual toward Gaussian (CLT). **The variance-gamma family contains Laplace as a special case ($\lambda = 1$), so the production calibration's "Laplace" result is consistent with the variance-gamma framework.**

1.4 Why this is actually a stronger result than v2

Variance-gamma residuals arise as variance-mixed Gaussians where the mixing measure is gamma:

$$\epsilon \,|\, V \sim \mathcal{N}(0, V), \quad V \sim \Gamma(\lambda, \theta) \quad \Longrightarrow \quad \epsilon \sim \mathrm{VG}(\lambda, \theta). \tag{1}$$

If $\lambda = 1$ (exponential durations), the marginal is Laplace. The *empirical fact* that compression-burst durations are $\Gamma(\text{shape} \approx 2.5)$ *predicts* a variance-gamma residual with $\lambda \approx 2.5$ (or modified by the burst-amplitude distribution; see §**??**).

The framework therefore predicts a measurable consistency:

$$\boxed{\lambda_{\mathrm{residual}} = (\text{gamma-shape of burst durations}).} \tag{2}$$

This is a self-consistency check that any future calibration can run as a one-line test.

2 Deterministic action: unchanged from v2

The connection-form Cholesky sector survives the empirical checks intact. We restate:

$$\omega_{\mathrm{1D}} = \alpha^2\,\mathrm{d}\alpha \wedge \mathrm{d}\beta, \qquad \mathcal{H}^{\mathrm{1D}}_{\mathrm{Ch}} = -\tfrac{1}{2}\,\alpha^2(\Sigma_{vv}(J,s) - \beta^2), \tag{3}$$

gauge weights $\alpha : 0$, $\beta : 1$, covariant time derivative $D_t^{(q)}\phi = \dot{\phi} + q\,(\partial_x u)\,\phi$. Hamilton equations reproduce dfmm exactly. Cold limit gives mass-coordinate dust action; Hessian degeneracy at $\gamma \to 0$ is the variational signature of rank collapse.

3 Stochastic dressing: variance-gamma from gamma-distributed bursts

3.1 Generalized compound-Poisson

Drop the assumption that burst durations are exponential. Instead: compression bursts arrive as a Poisson process at rate λ_b, each burst has random duration $T_b \sim \Gamma(k, \theta_T)$ and random intensity κ_b during the burst, and the noise variance accumulated in a save interval is

$$V = C_B^2 \sum_{n=1}^{N(T_{\mathrm{save}})} \kappa_n T_n. \tag{4}$$

At fixed compression intensity κ, V is a sum of gamma random variables, hence gamma-distributed. Including the multiplicative cascade of κ (She–Levêque structure) gives variance-gamma marginals.

Theorem (variance-gamma from variance mixing). If $\epsilon \,|\, V \sim \mathcal{N}(0, V)$ with $V \sim \Gamma(\lambda, \theta)$, then $\epsilon \sim \mathrm{VG}(\lambda, \theta)$ with density

$$f(\epsilon) = \frac{1}{\sqrt{\pi}\,\Gamma(\lambda)}\,\frac{1}{(2\theta)^{\lambda-1/2}}\,|\epsilon|^{\lambda-1/2}\,K_{\lambda-1/2}\Big(|\epsilon|/\sqrt{\theta}\Big), \tag{5}$$

where K_ν is the modified Bessel function of the second kind. Special cases: $\lambda = 1$ gives Laplace; $\lambda \to \infty$ (with θ scaled) gives Gaussian. Kurtosis is $3/\lambda$, monotonically decreasing in λ.

3.2 Updated MSRJD action

The stochastic-action contribution generalizes from quadratic vertex ($\propto \hat{\epsilon}^2$) to the variance-gamma cumulant generating function:

$$\mathcal{K}_{\mathrm{VG}}(\hat{\epsilon}; V_{\mathrm{typ}}) = -\lambda\,\log\!\big(1 - \theta\,\hat{\epsilon}^2/2\big), \tag{6}$$

which expands as

$$\mathcal{K}_{\mathrm{VG}} = \tfrac{1}{2}\,\lambda\theta\,\hat{\epsilon}^2 + \tfrac{1}{4}\,\lambda\theta^2\,\hat{\epsilon}^4 + \mathcal{O}(\hat{\epsilon}^6). \tag{7}$$

The leading term is the Gaussian noise vertex (recovering MSRJD at quadratic order), and the higher-order vertices encode the heavy tails. For the dfmm calibration with empirical $\lambda \sim 1$–2, only the quartic correction is numerically relevant; sextic and higher are negligible.

3.3 2D extension: per principal axis (preserved from v2)

Same prediction as v2: variance-gamma noise per compressive principal axis of the strain rate tensor S_{ij}, with axis-specific shape parameter λ_a set by the duration distribution of bursts along that axis. Pure shear has noise on one axis only; falsifiable.

4 2D matrix lift: principal-axis formulation

4.1 Why a direct tensor pairing fails

The natural conjecture from v2 was

$$\omega_{\mathrm{2D}}^{\mathrm{naive}} \stackrel{?}{=} (\det L_1)^2 \ \mathrm{Tr}(\mathrm{d}L_1 \wedge \mathrm{d}L_2^{\top}), \tag{8}$$

with L_1 the lower-triangular Cholesky of M_{xx} and L_2 the $(v|x)$ cross-block. Working out the 2D Liouville equations for $(a, b, c) \in L_1$, $(p, q, r, s) \in L_2$ (with $L_1 = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$) gives, schematically,

$$\dot{a} = p, \qquad \dot{b} = r + (c/a)q, \qquad \dot{c} = s - (b/a)q, \tag{9}$$

plus their L_2 counterparts. The off-diagonal entry q couples to b and c through (c/a) and (b/a) ratios that the naive $\mathrm{Tr}(dL_1 \wedge dL_2^{\top})$ pairing does *not* reproduce: Tr pairs a with p, b with r, c with s, leaving q unpaired and algebraic. But q is dynamical (it generates the cross-axis coupling). The naive pairing is therefore incomplete.

4.2 Principal-axis rotation as the fix

Cleaner approach. Work in the local principal-axis basis of M_{xx}: rotate to coordinates where L_1 is diagonal, $L_1 = \mathrm{diag}(\alpha_1, \alpha_2)$. The two principal axes carry independent 1D Cholesky structures, and cross-axis coupling enters via the Maurer–Cartan 1-form of the rotation $R(t) \in SO(2)$:

$$\omega_{\mathrm{2D}} \ = \ \alpha_1^2 \, \mathrm{d}\alpha_1 \wedge \mathrm{d}\beta_1 \ + \ \alpha_2^2 \, \mathrm{d}\alpha_2 \wedge \mathrm{d}\beta_2 \ + \ \omega_{\mathrm{rot}}(\alpha_a, \beta_a, \theta_R), \tag{10}$$

where θ_R is the rotation angle of the principal-axis frame and ω_{rot} is the Berry-curvature-like contribution. The first two terms are exactly two copies of the 1D weighted symplectic form, one per principal axis. The third term carries the cross-axis dynamics.

Status. The first two terms are derived; their reduction to 1D (set the second axis $\alpha_2 = \alpha_1$, $\beta_2 = \beta_1 = \beta$) recovers ω_{1D} correctly. The third term ω_{rot} is the natural Berry connection on the principal-axis bundle, but its explicit form requires further work. *This is the one piece of algebra I'm flagging as "derive in 2D paper, not before Milestone 1."*

Why this is the right structural choice.

1. **Reduction to 1D is automatic.** 1D scheme is a "single principal axis" specialization of the 2D action; no separate 1D code path needed.

2. **Noise structure aligns naturally.** The noise prediction (variance-gamma per compressive principal axis) lives on the same principal-axis decomposition as the symplectic form. The deterministic and stochastic sectors are both axis-diagonal in this frame.

3. **The rotation angle is a physical observable.** It tracks how the local M_{xx} ellipsoid orients itself relative to the strain eigenframe. In shell-crossing, the principal axes align with the caustic direction and ω_{rot} carries the rotation that brings L_1 to a singular limit.

4. 3**D extension is straightforward**: replace $SO(2)$ with $SO(3)$ and three principal axes; same structure with three diagonal 1D-form contributions plus a Berry curvature on $SO(3)$.

4.3 Verification of the 1D reduction

With one axis active ($\alpha_2 = 0$, $\beta_2 = 0$) and no rotation ($\theta_R = 0$), (**??**) collapses to $\omega_{2\text{D}} \to \alpha_1^2\, \mathrm{d}\alpha_1 \wedge \mathrm{d}\beta_1$, and the corresponding Hamiltonian $\mathcal{H}_{2\text{D}} \to -\frac{1}{2}\alpha_1^2\gamma_1^2$ recovers the v2 1D result exactly. ✓

5 Status of the four items

Item	Status	Notes
1. Drift decomp.	*flagged*	Cannot verify at small-data scale; production calibration needed. Framework accommodates any decomposition.
2. Burst durations	*falsified*	Gamma($\sim$2.5), not exponential. Stronger result: predicts VG residuals with same shape.
3. Sign-off	*given*	Connection form, Hessian-degeneracy interpretation, action structure all confirmed.
4. Matrix lift	*principal-axis form*	Naive Tr-pairing fails; principal-axis rotation gives clean structure. Berry-connection term flagged for 2D paper.

6 What changes for Milestone 1

1. Stochastic noise generator: implement variance-gamma (with shape parameter measured from burst statistics during calibration), with Laplace as a special case for backward compatibility.

2. Add a self-consistency check to the 1D scheme: at runtime, monitor both the burst-duration shape and the residual kurtosis; warn if they disagree.

3. Discrete action and EL equations on a 1D mass-coordinate mesh proceed as planned. The action is unchanged from v2 in the 1D specialization.

4. 2D extension uses (**??**); principal-axis rotation becomes a separate dynamical variable carried per cell. The Berry connection ω_{rot} is left for the 2D paper.

7 Honest caveats

1. All empirical fits are on a small ($N = 384$k) wave-pool sample. The production calibration uses a bigger ensemble and the Lagrangian-frame density disambiguation; my small-data fits should be re-run at production scale before publishing.

2. The variance-gamma derivation *predicts* a relation between burst-duration shape and residual kurtosis; the production calibration data is the right testbed.

3. The principal-axis Berry connection ω_{rot} in (**??**) is conjectural. The 1D reduction is verified.

4. I have not redone the variational consistency of the heat-flux constraint in 2D principal-axis basis; that's also for the 2D paper.

The deterministic action is solid. The stochastic dressing is upgraded from Laplace to variance-gamma with a single new parameter λ that's directly measurable. The 2D structure is sketched cleanly via principal-axis rotation, with one piece of algebra (Berry connection) flagged for later.

If this v3 is acceptable, I'd like to proceed to Milestone 1 (discrete action and 1D Rust implementation).